Tianxiang XIA

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EDUCATION

ETH Zürich, Zürich, Switzerland *since 09/2024*
Master's Degree, Major in Machine Intelligence GPA Major: **6/6** Overall: 5.87 (in progress)

Coursework Full marks (6/6) in Probabilistic AI, Deep Learning, Algorithms Lab, Optimization for Data Science, Big Data, Research in CS Memo, etc.

Sorbonne Université, Paris, France *09/2021 - 06/2024*
Double Bachelor's Degree in Mathematics & Computer Science 225 ECTS Grade: **19/20 (rank 1/70)**

Coursework - Mathematics: topology and multivariate calculus, functional analysis, linear and bilinear algebra, measure theory and probability, numerical analysis, algebra and arithmetic, etc.
- Computer Science: AI and operational research, numerical methods, OOP and functional programming, databases, algorithms, data structures, logic, OS, cryptography, networks, computer architecture, etc.

AWARDS AND HONORS

- ETH Zürich VIScon-19 Hackathon, as part of a three-student team, **First Prize** *10/2024*
- ICPC Southwestern Europe Regional Contest (SWERC 2023–2024, standings), **Bronze Medal** *01/2024*
- 30th China Team Selection Competition of International Olympiad in Informatics, **Silver Medal** *05/2018*

SKILLS AND RESEARCH INTERESTS

Programming languages C/C++, Python, Java, OCaml, LaTeX, SQL, Intel/ARM Intrinsics

Spoken languages Chinese Mandarin (native), English (IELTS C1), French (C1, Bachelor's program in French)

Technical skills AI (PyTorch, vLLM, LMCache), Unix-like Systems (Shell, Git, VIM), Docker, Web Development (HTML, CSS, JavaScript, React), Database Management (SQL, Spark, etc.)

Research Interests Optimization for large models and systems, geometries of world-model latent variables, etc.

WORK EXPERIENCE

Research Student at ETH Zürich, SRI lab, ODI lab (present) *since 03/2025*
Teaching Assistant at Algorithms Lab (algolab), ETH Zürich *09/2025 - 12/2025, 09/2026 - 12/2026*
- In charge of oral assessment for students' performance in Master's level algorithms course

Research Intern at Computing Product Lab, Huawei Zürich *06/2025 - 08/2025*
- R&D of high-hit-rate KV cache systems (report accepted as DAC 2026 WIP session poster) and online quantization for multi-LoRA agents and retrieval-augmented generation scenarios
- Contributed to open-source communities PTO-Kernels, LMCache-Ascend, CANN/ops-nn

Research Intern at Huawei Paris *06/2024 - 08/2024*
- Report (code) accepted for publication in *Annals of Mathematics and Artificial Intelligence*
- Designed and built a neural-symbolic reasoning engine for automatic Euclidean geometry theorem proving with human-computer interaction features through reasoning graph plotting and GeoGebra integration
- The project was continued by HarmonicMath and achieved gold-medal performance at IMO 2025

PROJECTS

Learning Structured Sparsity for Large Models: Bridging Model Compressibility and Efficient Inference
Report [1] accepted at AXIOM workshop, NeurIPS 2026; submitted to ICLR 2027 *03/2026 - 09/2026*
- Developed an ADMM-based optimization framework for jointly learning low-rank and hardware-friendly structured sparse representations during large language model pretraining
- Implemented an end-to-end pipeline with efficient low-rank proximal updates, a zero-shot elastic compression method, and custom TileLang/AVX2 inference kernels, cutting training overhead from 46% to 4% and advancing the empirical perplexity–memory–latency Pareto frontier

Hardware Optimization for Clifford Nets with report & code, at ETH Zürich *03/2025 - 06/2025*
- Optimized Clifford neural layers with AVX2 and ARM NEON intrinsics, achieving a 30% speedup over PyTorch
- Developed a generator for vectorized C kernels across Clifford-algebra dimensions, signatures, and CPU architectures

Automatic Neural Network Robustness Verification with code, at ETH Zürich *11/2024 - 12/2024*
- Implemented DeepPoly from scratch with learnable relaxations and obtained the highest evaluation score

Text Representation Analysis with slides, at LIP6, Sorbonne Université *02/2023 - 06/2023*
- Self-studied and applied topological data analysis (representation by similarity filtration with time skeleton, analysis by persistent homology) to essay assessment

References

[1] Tianxiang Xia, Kai Lion, Bernhard Schölkopf, Niao He, Michael Muehlebach, Melanie N. Zeilinger, and Hao Ma. Learning structured sparsity beyond SALAAD: Bridging model compressibility and efficient inference. In *AXIOM: Foundations of Efficient Deep Learning, NeurIPS Workshop*, 2026.